M3503 WEEK 4 2021

UNDETERMINED COEFFICIENTS: ANNIHILATOR APPROACH

Consider an nth order linear non-homogeneous equation with constant coefficients

$$a_n D^n y + a_{n-1} D^{n-1} y + \cdots + a_1 D y + a_0 y = f(x), \quad f(x) \neq 0 \quad \text{—①}$$

We rewrite equation ① as

$$L(y) = f(x),$$

where L denotes a linear nth order differential operator $a_n D^n + a_{n-1} D^{n-1} + \cdots + a_1 D + a_0$

We treat a differential operator D as an algebraic quantity so that if possible, L can be factored

eg $L(y) = D^2 y + 3Dy - 4y = (D^2 + 3D - 4)y = (D+4)(D-1)y$

ANNIHILATOR OPERATOR

If L is a linear differential operator with constant coefficients, and the function $f(x)$ is such that

$$L(f(x)) = 0,$$

then L is called the annihilator of $f(x)$.

Examples

① $D^3(x^2+2x+1) = D^2(2x+2)$

$= D(2)$

$= 0$

$\therefore D^3$ annihilates x^2+2x+1

In general, D^n annihilates the following functions and their linear combinations

$c, x, x^2, x^3, \ldots, x^{n-1}$,

where c is a constant and n is a positive integer

2. $(D-\alpha)^2 (xe^{\alpha x}) = (D^2 - 2D\alpha + \alpha^2)(xe^{\alpha x})$

$= D^2(xe^{\alpha x}) - 2\alpha D(xe^{\alpha x}) + \alpha^2 xe^{\alpha x}$

$= D(e^{\alpha x} + \alpha x e^{\alpha x}) - 2\alpha(e^{\alpha x} + \alpha x e^{\alpha x}) + \alpha^2 xe^{\alpha x}$

$= \alpha e^{\alpha x} + \alpha e^{\alpha x} + \alpha^2 x e^{\alpha x} - 2\alpha e^{\alpha x} - 2\alpha^2 x e^{\alpha x} + \alpha^2 x e^{\alpha x}$

$= 0$

$\therefore (D-\alpha)^2$ annihilates $xe^{\alpha x}$

In general, $(D-\alpha)^n$ annihilates the following functions and their linear combinations

$e^{\alpha x}, xe^{\alpha x}, x^2 e^{\alpha x}, \ldots, x^{n-1} e^{\alpha x}$

3. $[D^2 - 2\alpha D + (\alpha^2 + \beta^2)] e^{\alpha x} \cos(\beta x)$

$= D^2(e^{\alpha x}\cos(\beta x)) - 2\alpha D(e^{\alpha x}\cos(\beta x)) + (\alpha^2+\beta^2) e^{\alpha x}\cos(\beta x)$

$= D(\alpha e^{\alpha x}\cos(\beta x) - \beta e^{\alpha x}\sin(\beta x)) - 2\alpha(\alpha e^{\alpha x}\cos(\beta x) - \beta e^{\alpha x}\sin(\beta x)) + (\alpha^2+\beta^2) e^{\alpha x}\cos(\beta x)$

$= \alpha^2 e^{\alpha x}\cos(\beta x) - \alpha\beta e^{\alpha x}\sin(\beta x) - \alpha\beta e^{\alpha x}\sin(\beta x) - \beta^2 e^{\alpha x}\cos(\beta x)$

$-2\alpha^2 e^{\alpha x}\cos(\beta x) + 2\alpha\beta e^{\alpha x}\sin(\beta x) + \alpha^2 e^{\alpha x}\cos(\beta x) + \beta^2 e^{\alpha x}\cos(\beta x)$

$= 0$

$\therefore D^2 - 2\alpha D + (\alpha^2+\beta^2)$ annihilates $e^{\alpha x}\cos(\beta x)$

In general, $[D^2+2\alpha D+(\alpha^2+\beta^2)]^n$ annihilates the following functions and their linear combinations
$e^{\alpha x}\cos\beta x$, $xe^{\alpha x}\cos\beta x$, $x^2e^{\alpha x}\cos\beta x$, $\cdots$, $x^{n-1}e^{\alpha x}\cos\beta x$,
$e^{\alpha x}\sin\beta x$, $xe^{\alpha x}\sin(\beta x)$, $x^2e^{\alpha x}\sin\beta x$, $\cdots$, $x^{n-1}e^{\alpha x}\sin\beta x$

If L_1 annihilates y_1 and L_2 annihilates y_2, but $L_1(y_2)\neq 0$ and $L_2(y_1)\neq 0$, then

$$\begin{aligned} L_1L_2(c_1y_1+c_2y_2) &= L_1L_2(c_1y_1)+L_1L_2(c_2y_2) \\ &= c_1L_2L_1(y_1)+c_2L_1L_2(y_2) \\ &= c_1L_2(0)+c_2L_1(0) \\ &= 0 \end{aligned}$$

ANNIHILATOR APPROACH SUMMARIZED

Consider the differential equation

$$L(y)=f(x) \quad \text{---} \quad (1)$$

where L is an nth order differential operator with constant coefficients, and $f(x)$ is a linear combination of polynomials, exponentials, sines, cosines and their products.

1. Find the complementary solution i.e solve $L(y)=0$
2. Find the annihilator L_1 of $f(x)$ and apply it on both sides of equation (1) to get a homogeneous equation $L_1L(y)=0$
3. Find the solution to the homogeneous equation $L_1L(y)=0$. This solution is the general solution of $L(y)=f(x)$. Some terms from this solution will be duplicates of terms of the solution in step 1, y_p

is equal to all the terms that are not duplicates

4. Substitute y_p into the equation $L(y) = f(x)$ and determine the unknown coefficients

Examples

Solve

① $y'' - 9y = 54$

② $y'' + 4y' + 4y = 2x + 6$

③ $y'' - y' - 12y = e^{4x}$

④ $y'' + 25y = 6\sin x$

Sol^{n}

① $y'' - 9y = 54$

Auxiliary eq^{n}: $r^2 - 9 = 0$

$\Rightarrow r = \pm 3$

$\therefore y_c = c_1 e^{3x} + c_2 e^{-3x}$

$D(D^2 - 9)y = 0$

Auxiliary eq^{n}: $r(r^2 - 9) = 0$

$\Rightarrow r = 0, r = \pm 3$

$\therefore y = c_3 + c_1 e^{3x} + c_2 e^{-3x}$

$y_p = c_3$

$\Rightarrow y_p' = 0$

$\Rightarrow y_p'' = 0$

Substituting into the given equation, we get

$-9c_3 = 54 \Rightarrow c_3 = -54/9 = -6$

$\therefore y = -6 + c_1 e^{3x} + c_2 e^{-3x}$

② $y'' + 4y' + 4y = 2x + 6$

Auxiliary equation: $r^2 + 4r + 4 = 0$

$\Rightarrow (r+2)^2 = 0$

$\Rightarrow r = -2$

$\therefore y_c = c_1 e^{-2x} + c_2 x e^{-2x}$

$D^2(D^2 + 4D + 4)y = 0$

Auxiliary eqn: $r^2(r^2 + 4r + 4) = 0$

$r = 0$ or $r = -2$

$\therefore y = c_3 + c_4 x + c_1 e^{-2x} + c_2 x e^{-2x}$

$y_p = c_3 + c_4 x$

$\Rightarrow y_p' = c_4$

$\Rightarrow y_p'' = 0$

Substituting into the given equation, we get

$4c_4 + 4(c_3 + c_4 x) = 2x + 6$

equating coefficients, we get

x^0: $4c_4 + 4c_3 = 6$

x^1: $4c_4 = 2 \Rightarrow c_4 = \frac{1}{2}$

$\Rightarrow c_3 = 1$

$\therefore y = 1 + \frac{1}{2}x + c_1 e^{-2x} + c_2 x e^{-2x}$

③ $y'' - y' - 12y = e^{4x}$

Auxiliary eqn: $r^2 - r - 12 = 0$

$\Rightarrow (r-4)(r+3) = 0$

$\Rightarrow r = 4, r = -3$

$\therefore y_c = c_1 e^{4x} + c_2 e^{-3x}$

$$(D-4)(D^2-D-12)y=0$$

Auxiliary eqⁿ: $(r-4)(r^2-r-12)=0$

$r=4$, $r=-3$

$\therefore y = C_3xe^{4x} + C_1e^{4x} + C_2e^{-3x}$

$y_p = C_3xe^{4x}$

$y_p' = C_3e^{4x} + 4C_3xe^{4x}$

$y_p'' = 4C_3e^{4x} + 4C_3e^{4x} + 16C_3xe^{4x}$

$= 8C_3e^{4x} + 16C_3xe^{4x}$

Substituting into the given equation we get

$8C_3e^{4x} + 16C_3xe^{4x} - C_3e^{4x} - 4C_3xe^{4x} - 12C_3xe^{4x} = e^{4x}$

$\Rightarrow 7C_3e^{4x} = e^{4x}$

$\Rightarrow C_3 = \frac{1}{7}$

$\therefore y = \frac{1}{7}xe^{4x} + C_1e^{4x} + C_2e^{-3x}$

(4). $y'' + 25y = 6\sin x$

Auxiliary eqⁿ: $r^2+25=0$

$\Rightarrow r = \pm 5i$

$\therefore y_c = C_1\cos(5x) + C_2\sin(5x)$

$(D^2+1)(D^2+25)y=0$

Auxiliary eqⁿ: $(r^2+1)(r^2+25)=0$

$r=\pm i$, $r=\pm 5i$

$\therefore y = C_3\cos(x) + C_4\sin(x) + C_1\cos(5x) + C_2\sin(5x)$

$y_p = C_3 \cos x + C_4 \sin x$

$\Rightarrow y_p' = -C_3 \sin x + C_4 \cos x$

$\Rightarrow y_p'' = -C_3 \cos x - C_4 \sin x$

Substituting into the given equation, we get

$-C_3 \cos x - C_4 \sin x + 25 C_3 \cos x + 25 C_4 \sin x = 6 \sin x$

$\Rightarrow 24 C_3 \cos x + 24 C_4 \sin x = 6 \sin x$

$\Rightarrow C_3 = 0$ and $C_4 = 6/24 = 1/4$

$\therefore y = \frac{1}{4} \sin x + C_1 \cos(5x) + C_2 \sin(5x)$

2 Variation Of Parameters

This method can always be used to find a particular solution to a differential equation, provided the associated homogeneous equation can be solved.

Summarized Variation Of Parameters

Consider a second order non-homogeneous equation $ay'' + by' + cy = f(x)$ —— ①

To find a particular solution to equation ①

① . Find the complementary solution $y_c = C_1 y_1 + C_2 y_2$

② . Divide equation ① by a to get the form

$y'' + By' + Cy = g(x)$,

where $B = \frac{b}{a}$, $C = \frac{c}{a}$, $g(x) = f(x)/a$

(3) Compute the Wronskian $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$

and expressions W_1 and W_2, where

$$W_1 = \begin{vmatrix} 0 & y_2 \\ g(x) & y_2' \end{vmatrix}, \quad W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & g(x) \end{vmatrix}$$

(4) Integrate $u_1' = \frac{W_1}{W}$ and $u_2' = \frac{W_2}{W}$ to get u_1 and u_2, constants of integration may be taken as zero. The particular solution is $y_p = u_1 y_1 + u_2 y_2$

For higher-order differential equations

$$u_k' = \frac{W_k}{W}, \quad k = 1, 2, \ldots, n$$

$$W = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & & y_n' \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}, \quad W_1 = \begin{vmatrix} 0 & y_2 & \cdots & y_n \\ 0 & y_2' & & y_n' \\ \vdots & \vdots & & \vdots \\ g(x) & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix},$$

$$W_2 = \begin{vmatrix} y_1 & 0 & \cdots & y_n \\ y_1' & 0 & & y_n' \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)} & g(x) & & y_n^{(n-1)} \end{vmatrix}, \ldots, \quad W_n = \begin{vmatrix} y_1 & y_2 & \cdots & 0 \\ y_1' & y_2' & & 0 \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & g(x) \end{vmatrix}$$

Examples
Find the general solution to

① $y'' + y = \sec(x)$
② $y'' + 2y' + y = e^{-x}$
③ $3y'' - 6y' + 6y = e^x \sec(x)$
④ $y''' - 2y'' - y' + 2y = e^{4x}$
⑤ $y''' + y' = \tan(x)$

Soln

① $y'' + y = \sec(x)$

Auxiliary eqn: $r^2 + 1 = 0$

$\Rightarrow r = \pm i$

$y_c = C_1 \cos x + C_2 \sin x$

$$W(\cos x, \sin x) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$$

$$W_1 = \begin{vmatrix} 0 & \sin x \\ \sec x & \cos x \end{vmatrix} = -\tan x$$

$$W_2 = \begin{vmatrix} \cos x & 0 \\ -\sin x & \sec x \end{vmatrix} = 1$$

$u_1' = -\tan x$ and $u_2' = 1$

$\Rightarrow u_1 = \ln(\cos x)$ $\Rightarrow u_2 = x$

$\Rightarrow y_p = u_1 y_1 + u_2 y_2$

$= \cos(x) \ln(\cos x) + x \sin x$

$\therefore y = C_1 \cos x + C_2 \sin x + \cos x (\ln \cos x) + x \sin x$

② $y'' + 2y' + y = e^{-x}$

Auxiliary eq^n: $r^2 + 2r + 1 = 0$

$\Rightarrow (r+1)^2 = 0$

$\Rightarrow r = -1$

$\therefore y_c = c_1 e^{-x} + c_2 x e^{-x}$

$$W(e^{-x}, xe^{-x}) = \begin{vmatrix} e^{-x} & xe^{-x} \\ -e^{-x} & e^{-x} - xe^{-x} \end{vmatrix}$$

$$= e^{-2x} - xe^{-2x} + xe^{-2x}$$

$$= e^{-2x}$$

$$W_1 = \begin{vmatrix} 0 & xe^{-x} \\ e^{-x} & e^{-x} - xe^{-x} \end{vmatrix}$$

$$= -xe^{-2x}$$

$$W_2 = \begin{vmatrix} e^{-x} & 0 \\ -e^{-x} & e^{-x} \end{vmatrix}$$

$$= e^{-2x}$$

$u_1' = \frac{-xe^{-2x}}{e^{-2x}} = -x$, $u_2' = 1$

$\Rightarrow u_2 = x$

$\Rightarrow u_1 = -\frac{1}{2}x^2$

$\Rightarrow y_p = -\frac{1}{2}x^2 e^{-x} + x^2 e^{-x} = \frac{1}{2}x^2 e^{-x}$

$\therefore y = c_1 e^{-x} + c_2 x e^{-x} + \frac{1}{2}x^2 e^{-x}$

③ $3y'' - 6y' + 6y = e^x \sec(x)$

Auxiliary eqⁿ: $3r^2 - 6r + 6 = 0$

$\Rightarrow r^2 - 2r + 2 = 0$

$\Rightarrow r = \frac{2 \pm \sqrt{4 - 4(2)}}{2}$

$= 1 \pm i$

$\therefore y_c = c_1 e^x \cos x + c_2 e^x \sin x$

Rewriting the equation in standard form we get

$y'' - 2y' + 2y = \frac{1}{3} e^x \sec(x)$

$$W(e^x\cos x, e^x \sin x) = \begin{vmatrix} e^x \cos x & e^x \sin x \\ e^x \cos x - e^x \sin x & e^x \sin x + e^x \cos x \end{vmatrix}$$

$$= e^{2x}(\cos x \sin x + \cos^2 x - \cos x \sin x + \sin^2 x)$$

$$= e^{2x}$$

$$W_1 = \begin{vmatrix} 0 & e^x \sin x \\ \frac{1}{3} e^x \sec(x) & e^x \sin x + e^x \cos x \end{vmatrix}$$

$$= -\frac{1}{3} e^{2x} \tan x$$

$$W_2 = \begin{vmatrix} e^x \cos x & 0 \\ e^x \cos x - e^x \sin x & \frac{1}{3} e^x \sec(x) \end{vmatrix}$$

$$= -\frac{1}{3} e^{2x}$$

$u_1' = -\frac{1}{3} \tan x$, $u_2' = -\frac{1}{3}$

$\Rightarrow u_1 = \frac{1}{3} \ln(\cos x)$ $\Rightarrow u_2 = -\frac{1}{3} x$

$\Rightarrow y_p = \frac{1}{3} e^x \cos(x) \ln(\cos x) - \frac{1}{3} x e^x \sin x$

$\therefore y = c_1 e^x \cos x + c_2 e^x \sin x + \frac{1}{3} e^x [\cos(x) \ln(\cos x) - x \sin x]$

④ $y''' - 2y'' - y' + 2y = e^{4x}$

Auxiliary eqn: $r^3 - 2r^2 - r + 2 = 0$

$\Rightarrow (r-1)(r^2 - r - 2) = 0$

$\Rightarrow (r-1)(r-2)(r+1) = 0$

$\Rightarrow r = 1, r = 2, r = -1$

$\therefore y_c = c_1 e^x + c_2 e^{2x} + c_3 e^{-x}$

$$W(e^x, e^{2x}, e^{-x}) = \begin{vmatrix} e^x & e^{2x} & e^{-x} \\ e^x & 2e^{2x} & -e^{-x} \\ e^x & 4e^{2x} & e^{-x} \end{vmatrix}$$

$$= e^x \begin{vmatrix} 2e^{2x} & -e^{-x} \\ 4e^{2x} & e^{-x} \end{vmatrix} - e^x \begin{vmatrix} e^{2x} & e^{-x} \\ 4e^{2x} & e^{-x} \end{vmatrix} + e^x \begin{vmatrix} e^{2x} & e^{-x} \\ 2e^{2x} & -e^{-x} \end{vmatrix}$$

$= e^x[2e^x + 4e^x] - e^x[e^x - 4e^x] + e^x[-2e^x - e^x]$

$= e^x[6e^x + 3e^x - 3e^x]$

$= 6e^{2x}$

$$W_1 = \begin{vmatrix} 0 & e^{2x} & e^{-x} \\ 0 & 2e^{2x} & -e^{-x} \\ e^{4x} & 4e^{2x} & e^{-x} \end{vmatrix}$$

$$= e^{4x} \begin{vmatrix} e^{2x} & e^{-x} \\ 2e^{2x} & -e^{-x} \end{vmatrix}$$

$= e^{4x}[-e^x - 2e^x]$

$= -3e^{5x}$

$$W_2 = \begin{vmatrix} e^x & 0 & e^{-x} \\ e^x & 0 & -e^{-x} \\ e^x & e^{4x} & e^{-x} \end{vmatrix}$$

$$= e^{4x} \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix}$$

$= e^{4x}[-1 - 1]$

$= -2e^{4x}$

$$W_3 = \begin{vmatrix} e^x & e^{2x} & 0 \\ e^x & 2e^{2x} & 0 \\ e^x & 4e^{2x} & e^{4x} \end{vmatrix}$$

$$= e^{4x} \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix}$$

$$= e^{4x}[2e^{3x} - e^{3x}]$$

$$= e^{7x}$$

$u_1' = -\frac{1}{2}e^{3x}$, $u_2' = -\frac{1}{3}e^{2x}$, $u_3' = \frac{1}{6}e^{5x}$

$\Rightarrow u_1 = -\frac{1}{6}e^{3x}$ $\Rightarrow u_2 = -\frac{1}{6}e^{2x}$ $\Rightarrow u_3 = \frac{1}{30}e^{5x}$

$$\Rightarrow y_p = -\frac{1}{6}e^{4x} - \frac{1}{6}e^{4x} + \frac{1}{30}e^{4x} = -\frac{9}{30}e^{4x}$$

$$\therefore y = c_1e^x + c_2e^{2x} + c_3e^{-x} - \frac{9}{30}e^{4x}$$

⑤ $y''' + y' = \tan x$

Auxiliary eq^n: $r^3 + r = 0$

$\Rightarrow r(r^2+1) = 0$

$\Rightarrow r = 0, \; r = \pm i$

$\therefore y_c = c_1 + c_2\cos x + c_3\sin x$

$$W(1, \cos x, \sin x) = \begin{vmatrix} 1 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ 0 & -\cos x & -\sin x \end{vmatrix}$$

$$= \sin^2 x + \cos^2 x$$

$$= 1$$

$$W_1 = \begin{vmatrix} 0 & \cos x & \sin x \\ 0 & -\sin x & \cos x \\ \tan x & -\cos x & -\sin x \end{vmatrix}$$

$= \tan x \left[\cos^2 x + \sin^2 x\right]$

$= \tan x$

$$W_2 = \begin{vmatrix} 1 & 0 & \sin x \\ 0 & 0 & \cos x \\ 0 & \tan x & -\sin x \end{vmatrix}$$

$= -\sin x$

$$W_3 = \begin{vmatrix} 1 & \cos x & 0 \\ 0 & -\sin x & 0 \\ 0 & -\cos x & \tan x \end{vmatrix}$$

$= -\sin x \tan x$

$u_1' = \tan x$, $u_2' = -\sin x$, $u_3' = -\sin x \tan x$

$\Rightarrow u_1 = -\ln(\cos x)$ $\Rightarrow u_2 = \cos x$ $= \dfrac{-\sin^2 x}{\cos x}$

$= \dfrac{-[1 - \cos^2 x]}{\cos x}$

$= -\sec(x) + \cos x$

$= -\ln|\sec(x) + \tan x| + \sin x$

$\Rightarrow y_p = -\ln(\cos x) + \cos^2 x - \sin(x)\ln|\sec(x) + \tan x| + \sin^2 x$

$= -\ln(\cos x) - \sin(x)\ln|\sec(x) + \tan x|$

$\therefore y = c + c_2 \cos x + c_3 \sin x - \ln(\cos x) - \sin(x)\ln|\sec(x) + \tan x|$

CAUCHY-EULER EQUATIONS

These are equations of the form

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 x \frac{dy}{dx} + a_0 y = f(x)$$

To find the complementary solution, we try the solution $y = x^m$, where m is the solution to the auxiliary equation.

Consider the equation

$$a_2 x^2 y'' + a_1 x y' + a_0 y = 0$$

substituting $y = x^m$, we get

$$a_2 x^2 m(m-1) x^{m-2} + a_1 x m x^{m-1} + a_0 x^m = 0$$

$$\Rightarrow (a_2 m(m-1) + a_1 m + a_0) x^m = 0$$

$$\Rightarrow a_2 m(m-1) + a_1 m + a_0 = 0$$ which is the auxiliary equation

REAL AND DISTINCT ROOTS

If m_1 and m_2 are distinct real roots of the auxiliary equation, then

$$y = c_1 x^{m_1} + c_2 x^{m_2}$$

is the solution to the homogeneous equation.

REPEATED ROOTS

If $m_1 = m_2$ are roots of the auxiliary equation, then $y = c_1 x^{m_1} + c_2 x^{m_1} \ln(x)$ is the solution to the homogeneous equation

COMPLEX ROOTS

If $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$ are roots of the auxiliary equation, then

$$y = C_1 x^{\alpha} \cos(\beta \ln(x)) + C_2 x^{\alpha} \sin(\beta \ln(x))$$ is the solution to the homogeneous equation

Examples, solve

① $x^2 y'' - 2y = 0$

② $x^3 y''' - 6y = 0$

Soln:

① using $y = x^m$ yields the auxiliary equation

$m(m-1) - 2 = 0$

$\Rightarrow m^2 - m - 2 = 0$

$\Rightarrow (m-2)(m+1) = 0$

$\Rightarrow m = 2, \; m = -1$

$\therefore y = C_1 x^2 + C_2 x^{-1}$

② Inserting $y = x^m$ gives the auxiliary eqn

$m(m-1)(m-2) - 6 = 0$

$\Rightarrow m(m^2 - 3m + 2) - 6 = 0$

$\Rightarrow m^3 - 3m^2 + 2m - 6 = 0$

$\Rightarrow (m-3)(m^2+2) = 0$

$\Rightarrow m = 3, \; m = \pm i\sqrt{2}$

$\therefore y = C_1 x^3 + C_2 \cos(\sqrt{2} \ln(x)) + C_3 \sin(\sqrt{2} \ln(x))$

Examples

Use Variation of parameters to solve

① $x^2y'' - xy' + y = 2x$

② $x^2y'' + xy' - y = \ln x$

Sol^n ① Using $y = x^m$ gives the auxiliary eq^n

$m(m-1) - m + 1 = 0$

$\Rightarrow m^2 - 2m + 1 = 0$

$\Rightarrow (m-1)^2 = 0$

$\Rightarrow m = 1$

$\therefore y_c = c_1 x + c_2 x\ln(x)$

$W(x, x\ln x) = \begin{vmatrix} x & x\ln x \\ 1 & \ln x + 1 \end{vmatrix}$

$= x\ln x + x - x\ln x$

$= x$

Rewriting the eq^n in standard form we get

$y'' + \frac{1}{x}y' + \frac{1}{x^2}y = \frac{2}{x}$

$W_1 = \begin{vmatrix} 0 & x\ln x \\ 2/x & \ln x + 1 \end{vmatrix}$

$= -2\ln x$

$W_2 = \begin{vmatrix} x & 0 \\ 1 & 2/x \end{vmatrix}$

$= 2$

$u_1' = -\frac{2\ln x}{x}$, $u_2' = \frac{2}{x}$

$\Rightarrow u_1 = -(\ln x)^2$ $\Rightarrow u_2 = 2\ln x$

$\Rightarrow y_p = -x(\ln x)^2 + 2x(\ln x)^2$

$= x(\ln x)^2$

$\therefore y = c_1 x + c_2 x\ln(x) + x(\ln x)^2$

(2). Using $y = x^m$ gives the auxiliary eq^n

$$m(m-1) + m - 1 = 0$$

$$\Rightarrow (m-1)(m+1) = 0$$

$$\Rightarrow m = 1, \; m = -1$$

$$\therefore y_c = c_1 x + c_2 x^{-1}$$

$$W(x, x^{-1}) = \begin{vmatrix} x & x^{-1} \\ 1 & -x^{-2} \end{vmatrix}$$

$$= -x^{-1} - x^{-1}$$

$$= -2x^{-1}$$

Rewriting the eq^n in standard form we get

$$y'' + x^{-1} y' - x^{-2} y = x^{-2} \ln x$$

$$W_1 = \begin{vmatrix} 0 & x^{-1} \\ x^{-2} \ln x & -x^{-2} \end{vmatrix}$$

$$= -x^{-3} \ln x$$

$$W_2 = \begin{vmatrix} x & 0 \\ 1 & x^{-2} \ln x \end{vmatrix}$$

$$= x^{-1} \ln x$$

$$u_1' = \tfrac{1}{2} x^{-2} \ln x \qquad , \qquad u_2' = -\tfrac{1}{2} \ln x$$

$$\Rightarrow u_1 = \frac{-(1 + \ln x)}{2x} \qquad \Rightarrow u_2 = -\tfrac{1}{2}[x \ln x - x]$$

$$\therefore y = c_1 x + c_2 x^{-1} - \tfrac{1}{2}(1 + \ln x) - \tfrac{1}{2}(\ln x - 1)$$

$$= c_1 x + c_2 x^{-1} - \ln x$$

REWRITING CAUCHY-EULER EQUATIONS AS LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS

We make the substitution $x = e^t$ or $t = \ln x$

$$\frac{dy}{dx} = \frac{dy}{dt}\cdot\frac{dt}{dx}$$
$$= \frac{1}{x}\frac{dy}{dt}$$
$$= e^{-t}\frac{dy}{dt}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dt}\right)$$
$$= -\frac{1}{x^2}\frac{dy}{dt} + \frac{1}{x}\frac{d}{dx}\left(\frac{dy}{dt}\right)$$
$$= -\frac{1}{x^2}\frac{dy}{dt} + \frac{1}{x}\left(\frac{d^2y}{dt^2}\frac{1}{x}\right)$$
$$= -\frac{1}{x^2}\frac{dy}{dt} + \frac{1}{x^2}\frac{d^2y}{dt^2}$$
$$= \frac{1}{e^{2t}}\frac{d^2y}{dt^2} - \frac{1}{e^{2t}}\frac{dy}{dt}$$

Examples, solve

① $x^2y'' - xy' + y = 2x$

② $x^2y'' + xy' - y = \ln x$

Sol'ns

① let $x = e^t$, then $\frac{dy}{dx} = e^{-t}\frac{dy}{dt}$ and

$$\frac{d^2y}{dx^2} = e^{-2t}\frac{d^2y}{dt^2} - e^{-2t}\frac{dy}{dt}$$

Substituting into the eq^{n} we get

$$e^{2t}\left(e^{-2t}\frac{d^2y}{dt^2} - e^{-2t}\frac{dy}{dt}\right) - e^{t}\left(e^{-t}\frac{dy}{dt}\right) + y = 2e^{t}$$

$$\Rightarrow \frac{d^2y}{dt^2} - 2\frac{dy}{dt} + y = 2e^{t}$$

Auxiliary eq^{n}: $r^2 - 2r + 1 = 0$

$\Rightarrow (r-1)^2 = 0$

$\Rightarrow r = 1$

$\therefore y_c = c_1 e^{t} + c_2 t e^{t}$

$= c_1 x + c_2 x \ln(x)$

$$W(e^t, te^t) = \begin{vmatrix} e^t & te^t \\ e^t & te^t + e^t \end{vmatrix}$$

$= te^{2t} + e^{2t} - te^{2t}$

$= e^{2t}$

$$W_1 = \begin{vmatrix} 0 & te^t \\ 2e^t & te^t + e^t \end{vmatrix}$$

$= -2te^{2t}$

$$W_2 = \begin{vmatrix} e^t & 0 \\ e^t & 2e^t \end{vmatrix}$$

$= 2e^{2t}$

$u_1' = -2t$, $u_2' = 2$

$\Rightarrow u_1 = -t^2$ $\Rightarrow u_2 = 2t$

$\Rightarrow y_p = -t^2e^t + 2t^2e^t = t^2e^t$

$= (\ln x)^2 x$

$\therefore y = c_1 x + c_2 x \ln x + x(\ln x)^2$

② let $x = e^t$, then $\frac{dy}{dx} = e^{-t}\frac{dy}{dt}$

$$\frac{d^2y}{dx^2} = e^{-2t}\frac{d^2y}{dt^2} - e^{-2t}\frac{dy}{dt}$$

Substituting into the eq^n we get

$$e^{2t}\left(e^{-2t}\frac{d^2y}{dt^2} - e^{-2t}\frac{dy}{dt}\right) + e^{t}\left(e^{-t}\frac{dy}{dt}\right) - y = t$$

$$\Rightarrow \frac{d^2y}{dt^2} - \frac{dy}{dt} + \frac{dy}{dt} - y = t$$

$$\Rightarrow \frac{d^2y}{dt^2} - y = t$$

Auxiliary eq^n: $r^2 - 1 = 0$

$r = \pm 1$

$\Rightarrow y_c = c_1 e^t + c_2 e^{-t}$

$= c_1 x + c_2 x^{-1}$

$$W(e^t, e^{-t}) = \begin{vmatrix} e^t & e^{-t} \\ e^t & -e^{-t} \end{vmatrix}$$

$= -1 - 1$

$= -2$

$$W_1 = \begin{vmatrix} 0 & e^{-t} \\ t & -e^{-t} \end{vmatrix}$$

$= -te^{-t}$

$$W_2 = \begin{vmatrix} e^t & 0 \\ e^t & t \end{vmatrix}$$

$= te^t$

$u_1' = \frac{1}{2}te^t$, $u_2' = -\frac{1}{2}te^t$

$\Rightarrow u_1 = \frac{1}{2}(-te^{-t} - e^{-t})$, $u_2 = \frac{-1}{2}(te^t - e^t)$

$\Rightarrow y_p = \frac{1}{2}(-t-1) - \frac{1}{2}(t-1)$

$= -t$

$= -\ln x$

$\therefore y = c_1 x + c_2 x^{-1} - \ln x$